

SYSTEMATIC REVIEW

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Point-of-care testing for non-communicable diseases and health literacy outcomes: a systematic review

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Abstract

Background Health literacy responsiveness has been identified as a key action area in reducing the burden of non-communicable diseases (NCDs), which are a leading cause of death worldwide. Point-of-care (POC) testing has the potential to improve health literacy in individuals and communities. This systematic review aimed to synthesise the literature regarding POC testing for NCDs and health literacy outcomes.

Methods A systematic search was conducted across seven databases, Medline (via OVID), PsycINFO, Emcare, Scopus, ProQuest (Health databases), CINAHL, and Web of Science, between May 2018 and May 2023. Studies that investigated the utilisation of a POC device for measurement of NCD markers, and reported at least one outcome measure of health literacy in individuals over 10 years old were eligible for inclusion. Risk of bias in included studies was assessed using the Mixed Methods Appraisal Tool (MMAT). Results were mapped to the five WHO health literacy actions through narrative synthesis.

Results Thirteen studies were eligible for inclusion, primarily investigating POC assessment of diabetes, with some examining lipids or coagulation. Study cohorts were geographically limited to eight countries in predominantly high-income settings. Following POC testing relevant to diabetes and cardiovascular contexts, patients reported an enhanced understanding of health, motivation to make positive lifestyle changes, increased quality of life and decreased worry, increased confidence, and improved health markers.

Conclusion The findings from this review suggest that POC testing may support improvements in individual health literacy outcomes relevant to NCD management, specifically in diabetes and cardiovascular contexts. These findings reflect the potential of POC testing to contribute to more responsive health systems, by improving accessibility, results delivery, and communication of health information. However, the evidence base is limited in scope, and findings may not necessarily be generalisable. Future studies should prioritise examining a broader range of NCDs, particularly in underdeveloped regions, and employ specific validated health literacy tools.

Prospero registration CRD42022348011.

Keywords Point-of-care testing, POCT, Health literacy, Health education, Non-communicable diseases, Diabetes, Cardiovascular disease, Coagulation, Lipids, HbA1c

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Background

Non-communicable diseases (NCDs) are a major global health challenge, accounting for more than a third of the disease burden and 74% of all deaths [1, 2]. Of these deaths, more than three-quarters occur in low- to middle-income countries (LMICs) [2]. NCDs such as cardiovascular disease, diabetes, chronic respiratory diseases, cancer, and mental health conditions impose major health and economic burdens on individuals, communities, healthcare systems, and governments, and are therefore key targets for public health intervention [1]. NCD development is highly influenced by modifiable risk factors such as smoking, alcohol consumption, physical inactivity, diet quality, air pollution, and environmental stressors [1]. As such, interventions to reduce the burden of NCDs often focus on prevention via behavioural change [1].

Given the scale of the NCD burden, timely diagnosis and effective monitoring are critical. Point-of-care (POC) testing provides an accessible method of detecting and managing NCDs through the delivery of immediate results to inform clinical decision making [3]. POC testing is defined as diagnostic testing performed at or near the site of care, with results available during the patient encounter, and without the need for specimens to be sent to a central laboratory [3]. Results may then be communicated directly to patients, or via healthcare providers. For NCDs, POC testing options are available for monitoring cardiovascular disease markers (e.g. lipid profiles, international normalised ratio; INR), diabetes (e.g. HbA1c, blood glucose), and chronic respiratory diseases (e.g. exhaled nitric oxide; FeNO). Mental health conditions are not currently assessed via pathology-based POC diagnostics, and although there are emerging applications for cancer detection and monitoring (e.g. tumour markers), POC tests are not yet routinely implemented.

A fundamental advantage of POC testing is its capacity to bridge the gap in health equity by improving access and engagement [4–7]. Not only does POC testing benefit health access both geographically and culturally, it can also enhance patient engagement, through both the act of testing and the communication of results [7]. Importantly, how results are communicated, whether directly to patients or via healthcare providers, may influence how individuals understand and implement their health information.

Health literacy refers to the knowledge, confidence, and comfort of individuals and communities in accessing, understanding, appraising, remembering and using health-based information [8]. Earlier definitions focused on an individual's literacy and numeracy skills in healthcare contexts, but the concept has evolved to include both individual capacities and the organisational structures that support them [1, 9]. Poor health literacy has

been associated with increased hospitalisations, lower use of preventive measures such as vaccines, improper medication use, low health status, and higher mortality rates [10]. As an important determinant of health, poor health literacy can exacerbate existing health inequalities influenced by socioeconomic and cultural factors [11].

The World Health Organization (WHO) highlights five actions that demonstrate health literacy: *Access, Understand, Appraise, Remember, and Use* [8]. Individuals with adequate health literacy not only have the ability to *access, understand, and appraise* health information, but also *remember* to implement and *use* healthy skills and behaviours, such as taking medications or attending appointments [8]. These action domains provide the framework for this review. Beyond direct health literacy, related constructs such as activation, engagement, and empowerment also shape how individuals interact with health information. Patient activation refers to the knowledge, skills, and confidence in managing one's own health [12]; engagement reflects active involvement in healthcare processes [13]; and empowerment refers to gaining control over health decisions and actions [14]. Although these constructs are not direct measures of health literacy, they capture dimensions of confidence, knowledge, and motivation that overlap with the WHO health literacy actions.

While health literacy refers to individual capacities, health literacy responsiveness describes the ability of health systems and organisations to create environments that support engagement with health information and services. Health literacy responsiveness has been identified as a key action area in reducing the burden of NCDs [15]. The development and implementation of interventions and technologies that enhance health literacy responsiveness are therefore of great public health interest [16]. With its immediate patient involvement, POC testing has the potential to strengthen health literacy responsiveness by improving accessibility, communication, and patient-centred workflows [3]. Enhanced responsiveness can, in turn, support individual health literacy outcomes by improving understanding, confidence, and engagement with health information.

This systematic review therefore aimed to synthesise the current literature regarding POC testing and health literacy outcomes in relation to NCDs, including cardiovascular diseases, chronic respiratory diseases, and diabetes, in the context of the WHO health literacy framework.

Methods

Protocol and registration

The review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD42022348011). A full protocol

may be requested from the authors. Amendments to the original protocol included changes in author contributions, and health literacy outcomes were synthesised according to the World Health Organization (WHO) health literacy domains. The review was conducted and reported in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Additional file 1) [17].

Selection criteria

Records were eligible for inclusion if they met the following criteria:

- (i) the use of at least one POC testing device assessing NCD measures, including cardiovascular disease (e.g. lipid profiles, coagulation markers), chronic respiratory disease (e.g. FeNO), or diabetes (e.g. HbA1c, blood glucose). POC testing was considered across multiple contexts, including adult care, paediatric care, self-managed testing, clinic-based testing, or community health settings;
- (ii) reporting of at least one measure related to health literacy or health literacy responsiveness, including outcomes mapped to the five WHO health literacy actions (*Access, Understand, Appraise, Remember, Use*). This could include direct assessment of health literacy using validated instruments, or proxy measures such as patient activation, knowledge, awareness, confidence, satisfaction, engagement, behaviour change, or relevant clinical indicators;
- (iii) health literacy of individuals aged 10+ years, including older children and adolescents, with caregivers able to report their own health literacy outcomes based on POC assessment of a child;
- (iv) full-text peer reviewed journal article published within the five years prior to the search.

Studies were excluded if they were reviews of peer-reviewed studies.

The five-year limit was applied in line with WHO guidance to capture contemporary evidence on POC testing and health literacy, reflecting advances in diagnostic technologies and evolving definitions of health literacy. No restrictions on the type of study design, sample size, or language of manuscript were imposed.

Information sources

Records were identified by a research librarian and researcher (KM) through searches of electronic databases from 5 May 2018 to the date of the search, 5 May 2023. Databases searched included Medline (via OVID), PsycINFO (via OVID), Emcare (via OVID), Scopus, ProQuest (Health and Medicine databases only), CINAHL

(via EBSCO), and Web of Science (via Clarivate). Citation searches were not conducted.

Search strategy

The search strategy was developed in collaboration with an experienced research librarian. The search strategy consisted of MeSH terms including Point-of-Care Testing; Health Promotion; Health Knowledge, Attitudes, Practice; Patient Education as Topic; Health Literacy; Health Education; Life Style; Diet; Exercise; Health Behaviour; Smoking Cessation; Smoking Reduction; Tobacco Use Cessation; Alcohol Abstinence; Social Behaviour; and Social Participation. Keywords included domains of point-of-care (point of care, bedside, rapid, handheld, remote, mobile), testing (test, screening, diagnosis, monitor), patient (health, informed consent, disease, condition), knowledge (awareness, understanding, competence, comprehension), and behaviour (lifestyle, weight, diet, physical activity, alcohol, social, engage, community, smoking, tobacco).

The term 'review' was excluded by title and manuscript type. No filters were applied. A full search strategy by database can be found in Additional file 2.

Selection process

Resulting records from each database were imported into EndNote 20 where duplicates were provisionally removed. Records were then uploaded to Covidence systematic review management software [18] where a final duplication detection test was conducted. Titles and abstracts of resulting records were initially screened against eligibility criteria by two researchers (KM and LD), independently. Agreement between reviewers was high at 96.7%, with consensus required for further screening. Remaining records underwent independent full-text screening by the same two researchers (KM and LD), with consensus required for inclusion in the review. Agreement at full-text screening was 85.7%, with discrepancies resolved through discussion or, if necessary, adjudication by a third reviewer (MS).

Data collection

Results from eligible studies, including descriptive data, were extracted using a study-specific form created in Covidence [18]. Data extraction was completed independently by the same two researchers that screened the records, with discrepancies resolved by communication between the two researchers. Resulting data were then exported to Excel.

Data items

Data extracted from each included study consisted of: authors, publication year, country, dates study was undertaken, cohort description, participant demographics

(e.g. age, sex/gender, race), sample size, study design, data collection methods, exposure type (i.e. POC tests and measures), data analyses, and outcome data (i.e. health literacy measures). Health literacy measures could include direct assessment of health literacy, or indicators of health literacy via proxy measures such as patient activation, knowledge, motivation, adherence and behaviour change (validated or self-reported).

Risk of bias in individual studies

Risk of bias (quality) of individual studies was assessed using the Mixed Methods Appraisal Tool (MMAT) [19]. The MMAT assessed included studies based on the study design; qualitative only, quantitative (non-randomised, randomised, descriptive), or mixed methods. Risk of bias was assessed by two independent researchers, with any discrepancies resolved through discussion or a third reviewer. Studies were not allocated a rating based on recommendations from the MMAT author [19].

Data synthesis

Meta-analysis was not feasible due to heterogeneity in intervention type (e.g. community screening vs. clinical monitoring), disease focus (e.g. diabetes, cardiovascular), outcome measurement tools (e.g. Patient Activation Measure, PAM-13; behaviour change), and population characteristics, which restricted comparability. We therefore employed a framework-guided narrative synthesis. Qualitative and quantitative outcome data were extracted and deductively coded against the five WHO health literacy actions (*Access, Understand, Appraise, Remember, Use*) to enable thematic comparison. Each outcome was mapped to the domains that best reflected the construct measured, and findings were then categorised by condition and type of POC test.

Results

Study selection

The database searches resulted in the identification of 12,459 records (Fig. 1). After removing duplicates, 6,129 unique records were retained for screening (Fig. 1). Title and abstract screening resulted in 98 records for further screening. Of these, 11 reports could not be retrieved (abstract-only, $n=7$; non-written format, $n=2$; inaccessible, $n=2$), leaving 87 articles retained for full-text screening (Fig. 1). Thirteen articles were eligible for inclusion following the final screening stage (Fig. 1). This reduction highlights the limited availability of recent evidence investigating POC testing and health literacy outcomes.

Study characteristics

Thirteen studies were included in the review, with a total sample size of 21,265. Studies assessed POC testing exposures for management of diabetes [20–30], lipids [20,

30] and/or coagulation [29, 31, 32] (Table 1). Although chronic respiratory diseases were included in the search scope, no eligible studies assessing health literacy outcomes were identified. Across conditions, POC testing was implemented in varying contexts, including adult outpatient care [23, 24], paediatric diabetes management (with parental health literacy outcomes) [21, 22], at-home self-testing (e.g. INR testing) [31, 32], and community-based screening [27, 30].

Health literacy outcomes were reported through a range of proxy measures, including patient activation [25, 28], patient knowledge and experiences [21, 31, 32], Likert scale ratings of understanding [29, 30], qualitative interviews [20–24, 27], medication adherence [26, 31, 32], and behaviour change [21, 30]. As detailed in Table 2, measurement tools varied widely, including validated instruments such as PAM-13, clinical indicators (i.e. HbA1c, INR, and PDC), and study-developed questionnaires, knowledge scores, Likert scales, and qualitative interviews.

Included studies were conducted in eight countries: England, India, Ireland, Italy, New Zealand, South Korea, and the United States. The mean ages of adults assessed ranged from 31 to 70 years, with two studies assessing parental outcomes of children with type 1 diabetes [21, 22]. Basic demographic characteristics such as age and sex were consistently reported, however reporting of ethnicity was varied, with most studies not reporting data. Where reported, participants were predominantly European or non-Hispanic White [21–23, 28, 30], with limited representation of minority groups such as Asian or African ethnicity groups [23, 30]. One study included only Black men [27]. Other equity considerations, particularly socioeconomic status, were rarely reported, limiting synthesis across population subgroups. A summary of the included studies can be seen in Table 1.

Risk of bias within studies

Eight studies met all quality criteria assessed by the MMAT [20–26, 28]. The MMAT assessment showed consistently clear research questions and appropriate data collection across most studies. However, two quantitative studies did not meet criteria for the interpretation of results based on the data provided [27, 29], while two did not account for confounders in analyses [31, 32]. One descriptive study failed to meet the criteria of low non-response bias [30]. All results from the risk of bias assessment can be seen in Additional file 3.

Synthesis of results

To address the research question, six studies adopted a qualitative approach, six utilised quantitative methods, and one used mixed methods (Table 1). All outcomes were treated as proxy measures of health literacy or

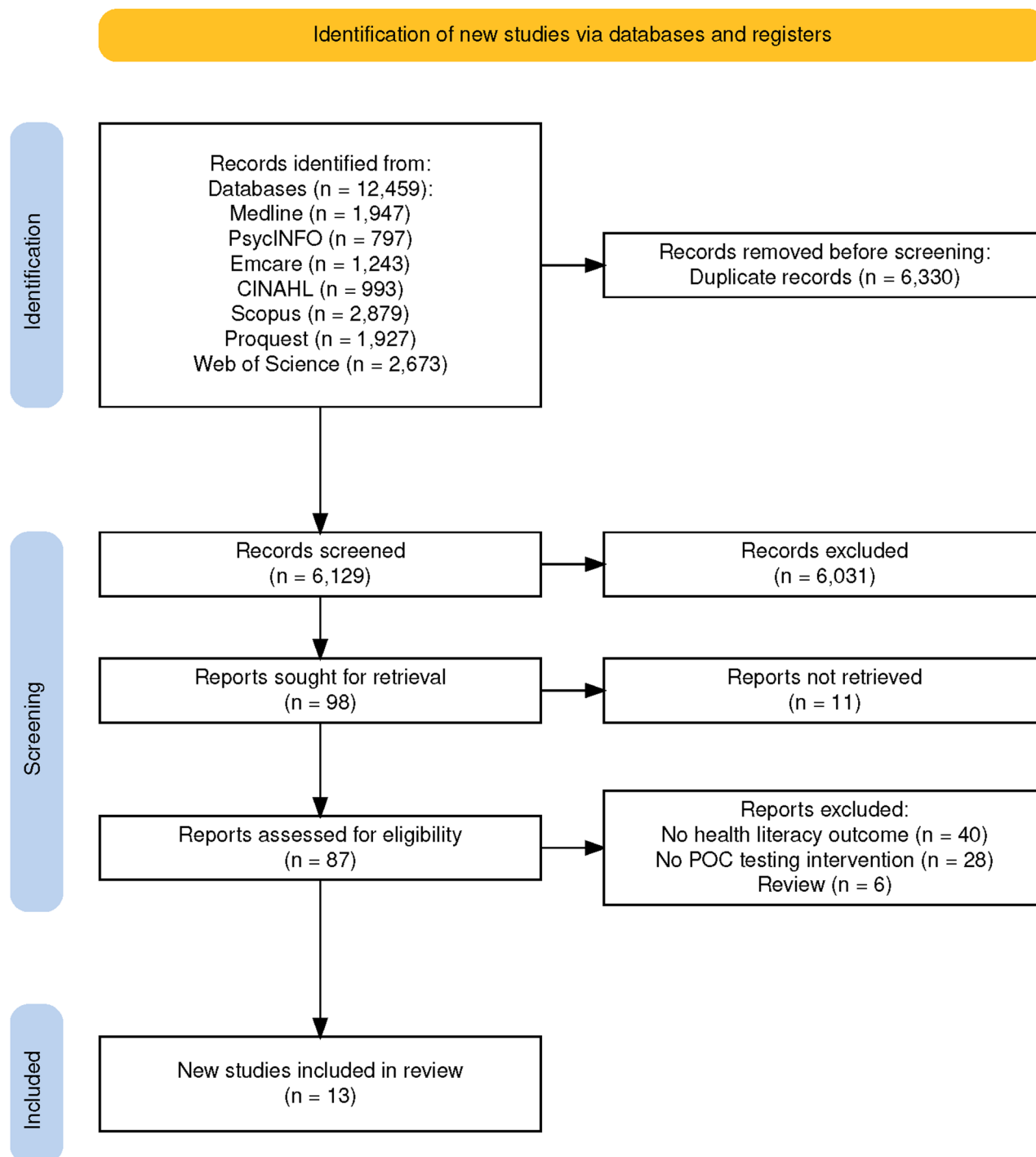


Fig. 1 PRISMA flowchart outlining the selection of studies

health literacy responsiveness, and were mapped to the five WHO health literacy actions (*Access, Understand, Appraise, Remember, Use*). Proxy outcomes and instruments used for each study are summarised in Table 2. Qualitative themes identified across studies, classified under the WHO domains, are presented in Table 3. Study-specific results are detailed in Additional file 4.

Access

Two qualitative studies reported participants felt more confident having assessment at the POC, as they were able to have an immediate discussion with the clinician after receiving results, which allowed them to receive advice and ask questions as they arose [20, 23]. Parents of children with type 1 diabetes reported that continuous

Table 1 Summary of study setting, population, point-of-care (POC) test type, sample size, demographics, and design across diabetes, coagulation, and mixed/other conditions

Study ID (Author, Year)	Country	Population	POC test	N	Sex (% male)	Age (mean $\pm$ SD/range)	Study Design
Diabetes – Type 1 (T1DM)							
Elbalsky, 2020 [21]	New Zealand	Parents of children with T1DM	CGM	12	8% (parents) 33% (children)	Median = 39 (29–47; parents) Median = 6.7 (1.2–14; children)	Qualitative interviews
Hilliard, 2019 [22]	United States	Parents of children with T1DM	CGM	55	12%	5.0 $\pm$ 1.5 (children)	Qualitative interviews
Diabetes – Type 2 (T2DM)							
Hirst, 2020 [23]	England	Clinic patients with diabetes	HbA1c	15	47%	56.9 (30–75)	Qualitative interviews
Michaud, 2018 [25]	United States	Patients with T2DM and recent hospital admission	Blood glucose	955	45%	60 (19–81)	Quantitative (PAM-13, HbA1c)
Munshi, 2021 [26]	United States	Adults with T2DM and health insurance	Blood glucose	18,006	43% (intervention) 42% (control)	54.2 $\pm$ 8.8 (intervention) 54.4 $\pm$ 9.1 (control)	Quantitative (PDC)
Rony, 2022 [27]	United States	Black men in barbershops (no prior diabetes diagnosis)	HbA1c	52	100%	42 (19–68)	Qualitative interviews
Su, 2019 [28]	United States	Patients with T2DM and recent hospital admission	HbA1c	1354	45%	59.6 $\pm$ 11.8	Quantitative (PAM-13, HbA1c)
Diabetes – Mixed (Type 1 & 2)							
Kang, 2022 [24]	South Korea	Diabetes clinic outpatients (58% T1DM, 42% T2DM)	CGM	19	37%	44.5 $\pm$ 13.7	Qualitative interviews
Coagulation							
Barcellona, 2018 [31]	Italy	Thrombosis centre patients on VKA therapy	INR	92	50%	64 $\pm$ 17	Quantitative (TTR, questionnaire)
Deepalakshmi, 2018 [32]	India	Patients on Warfarin	INR	110	74% (intervention) 89% (control)	61.4 $\pm$ 3.1 (intervention) 63.6 $\pm$ 2.9 (control)	Quantitative (INR)
Mixed/Other							
Butler, 2020 [20]	England	Adults with severe mental illness	HbA1c; lipid panel	483 (14 interviews)	60%	CHMT: 45 $\pm$ 11.8 (intervention); 43 $\pm$ 10.3 (control) EIS: 32 $\pm$ 12.2 (intervention); 31 $\pm$ 12.1 (control)	Qualitative interviews
Walkden, 2019 [29]	Ireland	Patients using remote telemonitoring	Blood glucose; INR	97	51%	70.2 $\pm$ 10.9 (HbA1c) 65.4 $\pm$ 1.9 (INR)	Mixed methods
Watson, 2020 [30]	England	Attendees from three <i>Health Melas</i> (fairs)	Blood glucose; cholesterol	375	41%	Not reported	Quantitative (questionnaire)

access to results supported their confidence in managing their child's condition [21, 22]. These outcomes were self-reported and reflect perceived improvements in access rather than independently verified changes.

Understand

Several qualitative studies of diabetes patients found that participants reported a greater understanding of their own [23, 24] or their child's [21, 22] physical health following POC testing interventions, supporting a reduction

in adverse events. Participants described improved comprehension of test results and clinical thresholds [20, 23]. Multiple studies also observed that this improved understanding extended to lifestyle behaviours in diabetes patients, such as diet, and the mechanisms through which they contributed to their physical health outcomes [22–24].

Considering quantitative results, Watson and Rajbhandari, et al. [30] reported that attendance at community *Health Melas* (Hindi term for “fair”), where an

Table 2 Proxy outcomes and instruments used for each study, classified under the World Health Organization (WHO) health literacy action domains (Access, Understand, Appraise, Remember, Use)

POC test	Study ID	Outcome (proxy type)	Tool	WHO health literacy action domain
CGM	Elbalshy, 2020	Awareness, quality of life, worry	Thematic interview	Access, Appraise, Use
	Hilliard, 2019	Awareness, knowledge, safety, worry	Thematic interview	Access, Appraise, Understand, Use
	Kang, 2022	Awareness, behaviour change, quality of life, understanding	Thematic interview	Access, Appraise, Understand, Use
HbA1c	Michaud, 2018	Activation, behaviour change	PAM-13 (validated); HbA1c (clinical)	Appraise, Understand, Use
	Hirst, 2020	Awareness, confidence, engagement, motivation, medication adherence	Thematic interview	Access, Appraise, Remember, Understand, Use
	Rony, 2022	Motivation	Thematic interview	Appraise
	Su, 2019	Activation, behaviour change	PAM-13 (validated); HbA1c (clinical)	Appraise, Understand, Use
Blood glucose	Munshi, 2021	Medication adherence	Proportion days covered (PDC)	Remember, Use
	Watson, 2020	Understanding, behaviour change	Likert scale questionnaire (study-developed)	Understand, Use
INR	Barcellona, 2018	Medication adherence, psychological impact, quality of life, satisfaction	TTR (clinical); questionnaire (study-developed)	Appraise, Remember, Understand, Use
	Deepalakshmi, 2018	Knowledge, medication adherence	INR (clinical); self-reported knowledge score (study-developed)	Remember, Understand, Use
Mixed	Butler, 2020	Engagement, understanding	Interview (study-developed)	Access, Remember, Understand
	Walkden, 2019	Empowerment, quality of life, satisfaction, worry	Likert scale questionnaire; interview (study-developed)	Appraise

Table 3 Summary of qualitative themes based on each WHO health literacy action

WHO health literacy action	Themes	Studies reported	N studies
Access	Immediate availability of results	Butler 2020, Elbalshy 2020, Hilliard 2019, Hirst 2020, Kang 2022	n = 5
	Enhanced discussions with clinicians	Butler 2020, Hirst 2020	n = 2
Understand	Greater understanding of lifestyle contributions to health outcomes	Hilliard 2019, Hirst 2020, Kang 2022	n = 3
	Improved comprehension of results and clinical thresholds	Butler 2020, Hilliard 2019	n = 2
Appraise	Reduced worry and improved quality of life	Elbalshy 2020, Hilliard 2019, Kang 2022, Walkden 2019	n = 4
	Increased motivation and empowerment	Hirst 2020, Kang 2022, Rony 2022	n = 3
	Enhanced confidence in managing health	Hirst 2020	n = 1
Remember	Encouraged to recall and clarify health information	Butler 2020, Hirst 2020	n = 2
	Recall medication schedules and instructions	Hirst 2020	n = 1
Use	Adoption of positive lifestyle behaviours	Elbalshy 2020, Hilliard 2019, Hirst 2020, Kang 2022	n = 4
	Improved clinical measures	Elbalshy 2020, Hilliard 2019, Kang 2022	n = 3
	Improved medication adherence	Hirst 2020	n = 1

overall health check was conducted, including random glucose and cholesterol POC testing, 98.4% of attendees reported an increased understanding of their health, and 93.1% were encouraged to make lifestyle changes [30]. As these events involved multiple health promotion activities in addition to POC testing, potential confounding must be considered when interpreting findings. A further two studies examined the change in HbA1c and PAM-13 scores following POC testing intervention, with PAM-13 scores increasing on average by 5.4 to 5.7 [25, 28]. Another study of patients on Warfarin found an average

of a 29% increase in self-reported knowledge scores following POC monitoring [32]. While PAM-13 is a validated instrument, other measures were author-developed or self-reported proxies, limiting comparability across studies.

Appraise

Improved quality of life and reduced worry was reported in four qualitative studies [21, 22, 24, 29], with two based on parental feedback [21, 22]. Other interviews described increased satisfaction, empowerment, and confidence

in managing health [23, 24, 27]. Among thrombosis centre patients on long-term vitamin K antagonist (VKA) therapy, 92.0% of those adopting an at-home self-testing intervention were satisfied with monitoring, compared to 39.0% under standard processes [31]. Intervention patients also reported greater comfort with side effects, and 78.0% indicated perceived improvements in quality of life [31].

Remember

In qualitative interviews, participants reported that POC testing encouraged them to recall and clarify health information during consultations [20, 23] and was associated with more consistent medication adherence [23] in diabetes contexts. These insights were supported by quantitative measures. Munshi and Amelung, et al. [26] reported that patients had a higher likelihood of remaining adherent to antidiabetic drugs compared to controls ($\beta = 1.4$, 95% CI = 1.3, 1.5), with adherence reflected in increased proportion of days covered (PDC) [26]. Barcellona and Mastino, et al. [31] found that patients using at-home INR testing achieved higher time in therapeutic range (TTR), ranging from 40.0% to 100.0% (median = 77.0%), compared to 24.0% to 89.0% (median = 71.0%) in controls. Similarly, another study reported improved INR control, with the proportion of patients in-range increasing from 77.5% pre-intervention to 87.3% post-intervention [32].

Use

Participants self-reported adoption of positive lifestyle behaviours, such as dietary changes and increased physical activity [21–24, 30]. At community *Health Melas* (fairs) 73.6% of attendees who had previously undergone a health check self-reported lifestyle changes afterwards [30]. Clinical outcomes showed reductions in high HbA1c levels (> 9%) from approximately 21% pre-intervention, to approximately 10% post-intervention [25, 28]. As described above, studies also reported associations with improved INR control [31, 32] and greater adherence to antidiabetic drugs [26]. While clinical measures support some findings, particularly in regards to medication adherence, lifestyle changes were self-reported and should be interpreted with caution.

Discussion

To our knowledge, this study is among the first to systematically review the literature on pathology-based POC testing for NCDs focusing on health literacy outcomes. Following systematic searches, thirteen studies conducted across eight countries were identified, investigating POC testing for diabetes/glucose, lipids, or coagulation management. POC testing is not yet widely available for the direct assessment of cancer or mental health conditions, and no eligible studies were identified

for chronic respiratory diseases. As such, the evidence base is restricted to diabetes and cardiovascular contexts, with most studies focused on diabetes. Included studies employed diverse methodologies (quantitative, qualitative, and mixed methods), but findings were generally consistent despite heterogeneity. Evidence suggested that access to POC testing may be associated with improvements across the WHO's five actions of health literacy: *Access, Understand, Appraise, Remember, and Use* health information.

Qualitative interviews consistently identified that *access* to health information was perceived as enhanced through the immediacy of POC test results [20–24]. Rapid feedback of results was reported to reduce delays and uncertainty, creating opportunities for patients to engage and communicate with clinicians more effectively [20, 24]. This was a consistent theme for parents of children with diabetes, who reported increased confidence in managing their child's condition [21, 22]. These findings suggest that the rapid nature of POC testing may be associated with supporting health literacy, not only by improving direct access to health information, but also by fostering more meaningful interactions between patients and health professionals.

Patients and parents using POC technologies for diabetes monitoring also reported enhanced *understanding* of their own [23, 24] or their child's [21, 22] health, including greater awareness of how lifestyle habits (e.g. diet) influenced outcomes [22–24], and improved comprehension of test results and clinical thresholds [20, 23]. These insights were complemented by quantitative proxy measures, including improvements in PAM-13 scores [25, 28], self-rated knowledge scores [32], and Likert scale assessments of understanding [30] and empowerment [29]. While PAM-13 is not a direct measure of health literacy, it is a validated tool that assesses an individual's knowledge, skills, and confidence in managing their health [12], aligning with WHO's actions of *appraising, understanding, and using* health information. Together, these findings suggest that POC testing may be associated with improved clinical understanding and reinforcement of the connection between health and lifestyle behaviours.

Appraising health information involves evaluating its relevance and implications for management [8], which was reflected through reports of reduced uncertainty and increased confidence in managing health. In qualitative studies, participants reported that POC testing for diabetes and coagulation reduced worry and improved quality of life [21, 22, 24, 29], while others described increased satisfaction, empowerment, and confidence in managing their health [23, 24, 27]. These findings were supported by quantitative improvements in PAM-13 scores [25, 28], and higher satisfaction and quality of life ratings [31].

In qualitative interviews, diabetes patients reported *remembering* and clarifying health information at the time of consultation when POC testing was used [20, 23]. POC testing in diabetes contexts was also associated with more consistent medication adherence [23]. These observations were reinforced by findings in coagulation monitoring, where associations with improved adherence was reflected in increased proportion of days covered (PDC) [26] and INR control [31, 32]. These results suggest that POC testing interventions may assist recall of health requirements, supported by measurable associations with improvements in treatment adherence.

The ability to *access, understand, appraise, and remember* health information appeared to be associated with the translation of knowledge into action through *use*. Across studies, participants reported adoption of positive lifestyle behaviours, such as dietary changes and increased physical activity [21–24, 30]. Quantitative data showed associations with improvements in medication adherence [26, 31, 32], and reductions in high HbA1c levels (> 9%) [25, 28]. Even a 1% decrease in HbA1c is associated with a significant reduction in diabetes-related complications, including cardiovascular disease, kidney damage, nerve damage, and eye problems [33, 34]. Collectively, these findings indicate that POC testing interventions may facilitate the translation of health information into action, supporting lifestyle change and improved disease management.

It is important to note that POC testing is rarely delivered in isolation, with communication of results also strongly influencing patient understanding and engagement. Mechanisms such as the immediacy of results, personalised provider explanation, and access to additional resources may reinforce knowledge, confidence, and comfort. Visible biomarker changes may strengthen the perceived link between behaviour and health outcomes, while integration into routine care can help normalise engagement and build confidence. In practice, POC testing was commonly delivered alongside broader care such as health consultations, community health events, and additional health measurements (e.g. blood pressure). These multi-layered interventions introduce several factors that may shape health knowledge and behaviour, and therefore, the observed outcomes cannot be attributed to POC testing alone. Rather, the findings highlight how POC testing, when combined with responsive communication and supportive care processes, may contribute to health literacy outcomes.

Strengths and limitations

This study was strengthened by the rigorous systematic review process conducted in line with the PRISMA statement [17]. Eligibility criteria were explicitly defined prior to commencing the literature search, with no restrictions imposed on language, region, or study type. The search

strategy was methodologically developed in collaboration with an experienced research librarian, following multiple pilot searches to refine database queries and MeSH terms across several databases to maximise coverage. Although comprehensive, the omission of device-specific and risk-factor terms may have limited retrieval of eligible studies, and citation searching was not undertaken. Future reviews should expand search terms to include specific NCDs, risk factors, and device-specific terminology to ensure broader coverage.

Study screening and data extraction were independently undertaken by two researchers, achieving high agreement, which improves reliability compared to single screening methods [35]. Included studies were limited to publications from the five years prior to the search, an intentional scope decision aligned with WHO guidance to capture contemporary evidence for POC tests and health literacy definitions. As the final search was conducted in May 2023, the study is limited in its consideration of more recent evidence or developments in POC testing, and the findings should be interpreted as reflecting the evidence base up to that date. Although newer studies may now exist, findings remain highly relevant and provide a valuable foundation for understanding the role of modern POC testing in health literacy. Additionally, 11 full-text reports were unable to be retrieved, largely due to being abstract-only or presented in non-written formats. While these reports may lack sufficient methodological detail for full appraisal, their exclusion may still introduce retrieval bias and contribute to the narrow evidence base identified.

Meta-analysis was not feasible due to variability between interventions, disease focus, measurement tools, and population characteristics. Outcomes were assessed using diverse instruments, ranging from validated tools (e.g. PAM-13) to study-developed questionnaires, qualitative interviews, and clinical markers of adherence or control. This mix of validated and self-reported proxy measures introduces differences in measurement rigour, further limiting comparability across studies. Despite this, shared self-reported proxy outcomes such as patient knowledge, confidence, engagement, and behaviour change enabled narrative synthesis via the framework of WHO health literacy actions. Designs were predominantly cross-sectional or short-term, with limited longitudinal follow-up. Furthermore, equity indicators were inconsistently reported, limiting disaggregated analysis. Although age and sex variables were consistently reported, ethnicity and socioeconomic indicators were often absent, restricting the ability to generalise findings to diverse populations. These limitations highlight the need for more consistent measurements using validated health literacy instruments, longer follow-ups, and systematic equity reporting in future work.

While risk of bias assessment using the MMAT demonstrated generally strong methodological quality across the included studies, several limitations were also identified. Several studies did not adequately account for potential confounders, making it difficult to isolate the contribution of POC testing. Two studies did not meet the criterion for appropriate interpretation of results based on the data provided [27, 29], however this was addressed by considering only the raw outcome data in the synthesis. One study did not meet the criterion for low non-response bias [30], potentially affecting the representativeness of its findings. Although these issues were present in a minority of studies, they may reduce confidence in some reported associations and should be considered when interpreting the overall strength of the evidence base.

Knowledge gaps and future directions

This review identified a small number of recent studies that investigated NCD-related POC testing and health literacy. POC measurements included diabetes/glucose, lipid panels, and coagulation, with the majority conducted in populations with diabetes. Findings therefore are not generalisable across all NCD contexts. Cohorts were geographically limited to eight countries, with only one of the thirteen included studies representing a LMIC (India). No studies were identified from Africa, which is a major limitation given the disproportionate burden of NCDs in LMICs [2]. In LMICs, POC testing has particular potential to reduce inequities in the diagnosis, and subsequent prevention, treatment, and care of NCDs. In settings with limited resources, household funds can be quickly depleted by NCD care and treatment, furthering socioeconomic inequities [2]. The introduction of high-impact interventions tailored for implementation in LMICs is therefore imperative. Expanding future research to include diverse geographies (e.g. LMICs), as well as broader POC tests (e.g. chronic respiratory disease), will strengthen understanding and generalisability, particularly in priority populations.

Although some studies utilised validated Patient Activation Measures or assessment of biomarker changes, no identified studies applied specific health literacy measurement tools for outcome assessment. As a key determinant of health, future studies should adopt the use of global health literacy instruments, such as the Health Literacy Questionnaire (HLQ) [36], to generate more targeted data and improve between-study comparisons. Furthermore, longitudinal designs and clarity in reporting equity indicators would benefit future research. Future work may also consider how system-level implementation and responsiveness in scaling up POC testing influence individual health literacy outcomes.

Conclusions

The findings from this review suggest that POC testing for NCD measures, specifically in diabetes and cardiovascular contexts, may support health literacy outcomes. POC measures were limited to lipids and coagulation, and primarily blood glucose and HbA1c. Reported interventions were associated with proxy measures of health literacy, including an enhanced understanding of health, motivation to adopt positive lifestyle changes, greater medication adherence, increased confidence, and improved health markers. Despite this, current research in LMICs, where the burden of NCDs is particularly high, is limited. To enhance this understanding, further studies should prioritise the relationship between health literacy and POC testing for NCDs in underdeveloped regions.

Abbreviations

CGM	Continuous glucose monitoring
FeNO	Exhaled nitric oxide
HbA1c	Haemoglobin A1c
HLQ	Health Literacy Questionnaire
INR	International normalised ratio
LMICs	Low- to middle-income countries
MMAT	Mixed Methods Appraisal Tool
NCDs	Non-communicable diseases
PAM-13	Patient Activation Measure
PDC	Proportion of days covered
POC	Point-of-care
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-analyses
TTR	Time in therapeutic range
T1DM	Type 1 diabetes mellitus
T2DM	Type 2 diabetes mellitus
VKA	Vitamin K antagonist
WHO	World Health Organization

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-026-26519-w>.

Additional file 1: (.docx): PRISMA checklist Completed PRISMA 2020 Checklist and Abstract Checklist for reporting on systematic reviews, including page numbers and table/figure numbers referencing where the guideline has been addressed (if available).

Additional file 2: (.docx): Search strategy Systematic review search strategy for each database.

Additional file 3: (.docx): Risk of bias assessment Risk of bias assessment within studies included in the systematic review, utilising the Mixed Methods Appraisal Tool (MMAT).

Additional file 4: (.docx): Table of results Results for studies included in the systematic review.

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Disclaimer

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Authors' contributions

KM participated in the design of the study, screened studies for selection, extracted the data, performed data synthesis, and drafted the manuscript. LD contributed to the design of the study, study screening and data extraction. TR participated in the design of the study and drafting of the manuscript. SM contributed to the study conception and critically revised the manuscript. MS and IG participated in the conception and design of the study, and supervised drafting of the manuscript. All authors have read and approved the final version of the manuscript.

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Data availability

The datasets used during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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